Reinforcement Learning (DSAI 402)

Lecture 2: RL Basics

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Agenda

Objectives: you should be able to name the state, action, and reward of any repeating decision, and explain why the future matters more than the next step alone.

- Play: a short game, no vocabulary yet
- Name it: agent, environment, state, action, reward
- Formalize it: the loop, return, discounting, the Markov property
- Apply it: the first application paper, contextual bandits for news

The Treasure 3x4 Grid

- The team starts at 🟢. Each turn, pick a direction: **Up / Down / Left / Right**.
- Somewhere on this grid: **one treasure (+10)**, a few **traps (−5)**, everything else is **0**.
- You don't get the map. You only find out what's on a square once you step on it.
- You have 5 moves. The game ends when you reach the treasure or run out of moves.
- Traps cost −5 each time and don't end the game. Hitting the edge wastes a move and scores 0.

	Col 1	Col 2	Col 3	Col 4
Row 1				
Row 2	🟢			
Row 3				

Playing It

Turn	Path	Landed on	Running total
1	F F R F L	Yes	-5 + 0 + -5 + 0 + 10
2	R F L F F	Yes	0+ 0 + -5 + 0 + 10
3	F R F F L	Yes	-5 + 0 + -5 + 0 + 10

Debrief

- Each turn, what were you choosing?
- What did that choice depend on?

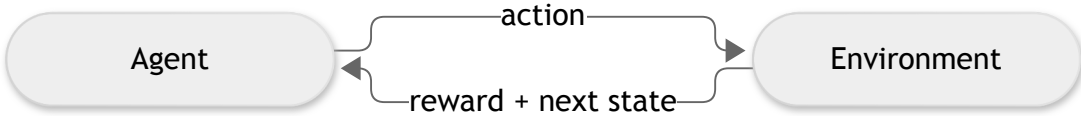
- What did you get back after each move?
- What number did you actually care about — one move's points, or your running total?

- Would it matter if the treasure showed up sooner rather than later?

- Did the squares you visited two turns ago change what you should do now?

You Already Used All Five

Now let's name them properly.



- **Agent** — the thing making decisions
- **Environment** — everything the agent interacts with
- **State** — the situation it's in
- **Action** — the choice it makes
- **Reward** — the feedback it gets back

Agent and Environment

Agent

The thing making decisions.

- Game: your team

Environment

Everything the agent interacts with, and reacts to its actions.

- Game: the grid, the traps, the treasure

State (S_t)

The situation at time t, i.e. the facts that decide what can happen next.

- Game: your square and how many moves are left

Action (A_t)

A choice made in a given state.

- Game: Up / Down / Left / Right

Reward (R_{t+1})

The number the environment sends back after action A_t.

- Game: +10 treasure, −5 trap, 0 otherwise

Important

A reward is feedback on one step, not a verdict on the whole episode. Confusing the two is the most common beginner mistake.

The Interaction Loop

Start in a state, take an action, get a reward, land in a new state — repeat.

$$S_0 \xrightarrow{A_0} R_1, S_1 \xrightarrow{A_1} R_2, S_2 \xrightarrow{A_2} R_3, S_3 \ \ldots$$

This is exactly what teams were doing, turn after turn. Every method in this course is a variation on this loop.

Risky Path vs. Safe Path

Risky — 3 moves

	Col 1	Col 2	Col 3	Col 4
Row 1				
Row 2	🟢 0	💥 1	· 2	⭐ 3
Row 3				

Right, Right, Right — straight through the trap.

Reward per step: $-5, 0, +10$

Safe — 5 moves

	Col 1	Col 2	Col 3	Col 4
Row 1	1	2	3	4
Row 2	🟢 0	·	·	⭐ 5
Row 3				

Up, Right, Right, Right, Down — around both traps.

Reward per step: $0, 0, 0, 0, +10$

Return

Return (G_t): the total reward collected from time t onward — not just the next step.

$$G_t = R_{t+1} + R_{t+2} + R_{t+3} + \dots$$

- Risky path (through the trap): $-5 + 0 + 10 = 5$
- Safe path (around it): $0 + 0 + 0 + 0 + 10 = 10$

> **Tip**
> The agent's goal is to maximize *return*, not any single reward. A tempting +10 now can be a bad idea if it costs you a −20 later.

Discounting

We usually care more about reward that arrives sooner. In the game, sooner only mattered because you could run out of moves. Discounting builds this preference in on purpose.

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots, \quad 0 \leq \gamma \leq 1$$

Important

A small reward that comes soon can beat a larger reward that comes late. With $\gamma = 0.9$, +10 after 1 move beats +12 after 10 moves, since $12 \times 0.9^9 \approx 4.6$.

Markov Property

To predict what a move does, do you need your whole history, or just your square, your moves left and the move itself?

$$P(S_{t+1}, R_{t+1} \mid S_t, A_t, S_{t-1}, A_{t-1}, \ldots, S_0, A_0) = P(S_{t+1}, R_{t+1} \mid S_t, A_t)$$

The current state already summarizes everything relevant from the past. In our grid the state is your square and your moves left. What you found out about the hidden map is not part of the state. Learning the map over many games is the job of the RL algorithm.

Important
Most methods in this course lean on this assumption. When it fails, we fix it by putting the parts of the past that matter into the state. For example, DQN (week 10) stacks the last 4 game frames into its state.

Markov Process

A sequence of states with no actions, where the next state depends only on the current one:

$$P(S_{t+1} \mid S_t, S_{t-1}, \ldots, S_0) = P(S_{t+1} \mid S_t)$$

It is described by:

- A set of states S
- A transition matrix P (probability of moving from one state to another)

Example: weather, one state per day

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S S R S S S R R S R
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The treasure grid becomes a Markov process once we fix a policy that picks each move from the current state alone and never changes. The teams did not play this way, since they changed plans after each discovery.

Practice Check

Identify the agent, environment, state, action, and reward:

Scenario: a thermostat that learns your comfort preferences

- It watches the room temperature and time of day
- It can raise or lower the target temperature
- You adjust it back down if it’s too warm; you leave it alone if it’s comfortable

Agent

The decision-maker

Environment

Everything it interacts with

State S_t

The situation at time t

Action A_t

A choice made in a state

Reward R_{t+1}

Feedback after one step

Return G_t

Total reward from now on

Discount γ

How much future reward is worth today

Markov property

The state summarizes the past

Before We Meet Again

1. **Write a short response to the Li et al. paper** — see Assignment 2 for the exact prompt.
2. **Identify agent, state, action, reward** for a problem of your choice.
3. Next lecture: the Markov Decision Process — policy, value functions, Bellman equations, and the chip-floorplanning paper.

> **Tip**
> If you're stuck naming the state for your project idea, that's normal — it's usually the hardest part to get right, and it's exactly what week 3 gives you tools for.

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